BASEL III Market Risk Modelling Using Monte Carlo VaR

Regulatory Framework
- Basel III
- BCBS Guidelines
- Internal Model Validation

Model Limitations
- Linear assumption
- Normality assumption
- Tail Risk underestimation

Factor Diagnostics
- R^2 Model Fit
- p-value significance
- Variance Inflation Factor (VIF)

Factor-Based Monte Carlo VaR
CAPM-Style Model
XLK, MTUM, VLUE
Multivariate Normal

Return-Based Monte Carlo VaR
- Normal Distribution
- Mean & Volatility
- 95% 1-Day VaR

Probability Space:
Ω : Outcomes
E : Events
P : Probability Measure
X : Ω → R

Random Variables

VaR Methodologies

Normal Var:
$VaR_\alpha = -(\mu + z_\alpha \sigma)$

Simulated Portfolio Returns
$R^* = \beta^T f^*$

Definition of VaR
$VaR\alpha = \inf\{x : FL(x) \geq \alpha\}$

Confidence level:
α = 95%

Uses mean & volatility

F_t ~ N(μ_F , Σ_F)

Monte Carlo Simulation
- Random Sampling
- 10,000 Simulations
- Loss Distribution
- Tail Risk (5%)

F_1t : Tech Factor Return
F_2t : Momentum Factor Retur
F_3t :Value Factor Return

Systematic Risk factor returns

Portfolio Weights
AAPL 30%
MSFT 25%
NVDA 20%
GOOG 15%
AMD 10%

CDF OF LOSS
$FL(x) = P(L \leq x) = \Phi((x - \mu_i) / \sigma_i)$
$\Phi(z) = \int$ from $-\infty$ to z $[1 / \sqrt{(2\pi)}]\, e^{(-u^2/2)}\, du$

Generates empirical loss distribution

PDF:
$f_L(x) = 1 / \text{sqrt}(2\pi\sigma_L^2) \cdot \exp(-(x - \mu_L)^2 / (2\sigma_L^2))$

are treated as random variables to capture uncertainty in market

Portfolio Returns

are generated via Monte Carlo simulation to estimate the loss

Simulated (future) portfolio returns

Defines Portfolio for

Portfolio Construction
- AAPL, MSFT, NVDA
- GOOG, AMD
- Fixed Weights
- Tech Equity Portfolio

Depends On

$\varepsilon : \Omega \to R$

Idiosyncratic Risk

computes return for

Return Calculation
- Adjusted Close Prices
- Log Returns
- Portfolio Returns

Train-Test Split
- 80% Training Data
- 20% Testing Data

$R_t = \beta^T F_t + \varepsilon_t$

Expanded form

$R_t = \beta_1 F_{1,t} + \beta_2 F_{2,t} + \beta_3 F_{3,t} + \varepsilon_t$

$L_t = -R_t$

Portfolio Loss

parametrized by

Statistical Estimation
Mean
Volatility
Covariance Matrix

Factor Moments:
$\mu_F = E[F_t], \Sigma_F = Cov(F_t)$

Variance Decomposition
Systematic Risk
Idiosyncratic Risk

Validates

Backtesting & Validation

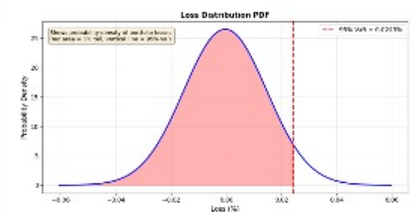
System Pipeline

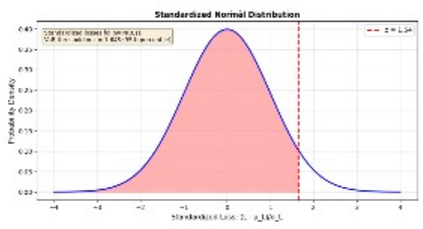
Distributional Assumptions:
- Factor returns vector:
F_t ~ N(mu_F, Sigma_F)
- Mean of factors:
mu_F = E[F_t]
- Covariance matrix of factors:
Sigma_F = Cov(F_t) = E[(F_t - mu_F)(F_t - mu_F)^T]
- Distribution of Portfolio Return: R_t = β^T F_t + ε_t
R_t ~ N(β^T mu_F , β^T Sigma_F β + sigma_epsilon^2)
- Mean of portfolio return: mu_R = β^T mu_F
- Variance of portfolio return: sigma_R^2 = β^T Sigma_F β + sigma_epsilon^2

Exception Rate Analysis
Compare predicted VaR with realized portfolio loss

- Historical Backtesting
- Rolling Backtesting (250 Days)
- VaR Exceptions

CSV Market Data
Train Test Split
Portfolio Construction
Loss Distribution
Backtesting



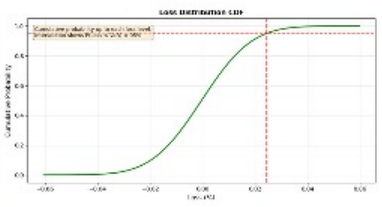


Is Evaluated Using

Basel Traffic Light
- Green (0–4)
- Yellow (5–9)
- Red (10+)

Model Evaluation
Strengths:
- Interpretable
- Efficient

Limitations:
- Gaussian assumption
- Tail risk underestimation



Ensures Compliance With

Ensures Basel III Compliance

Monte Carlo VaR — Randomized Core

10,000 simulations · Empirical loss distribution · Worst 5% → VaR estimate · seed=42 for reproducibility · Replaces analytic formula
Hybrid: Probabilistic scenarios + Fixed decision rule (VaR = 95th percentile of L_i) · $R_i \sim N(\mu,\sigma)$ i=1…10K · $L_i = -R_i$ (for $R_i < 0$)

Plan A: Normal Distribution
np.random.normal(μ, σ, 10000)
Primary method · if ≥ 100 simulated losses
Works for 95% of typical trading days
LLN: 10K → matches real distribution

Plan B: Bootstrap Sampling
Samplingnp.random.choice(hist_losses, n)
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Factor-Based MC (Alt. Method)
F_t ~ N(μ_F, Σ_F) multivariate
R* = β'f* → simulate factor returns
Captures inter-factor correlations
Distinct from return-based A/B/C

Probabilistic Assumptions
1. LLN: 10,000 sims → distribution converges to truth
2. Bootstrap: historical losses predict future losses.
3. EVT: rare events follow non-normal patterns.
4. Normal Dist.: valid for ~95% of trading days.
Each Plan (A/B/C) implements one assumption above

Why Randomization?
Fat tails: crashes happen more often
Formula assumes perfect bell curve
MC gives range of outcomes → realistic
LLN converges at 10,000 simulations
Market is volatile - not predictable

Hybrid Model Logic
RANDOM: generate 10,000 return scenarios
DETERMINISTIC: VaR (95%) = 5th percentile of losses
Random inputs + fixed rule → consistent VaR
Hybrid approach → reliable without false precision.

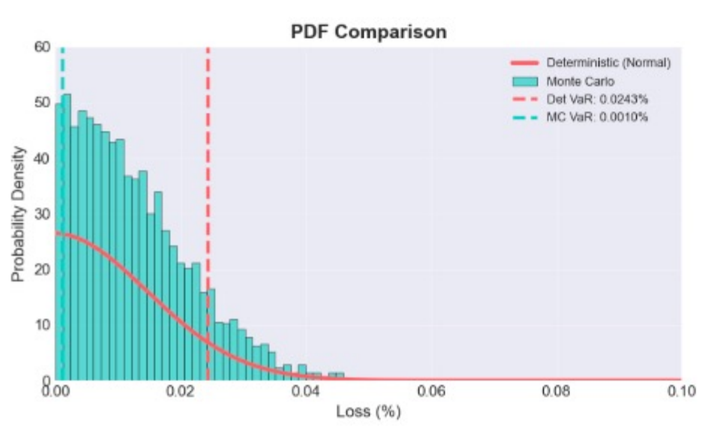
Sources of Uncertainty
1. Parameter: limited historical data (~400 days)
2. Distribution: fat tails (non-normal returns)
3. Sampling: 10k simulations → VaR variation
4. Model: factors may miss some risks
5. Seed (42): reproducible, not identical

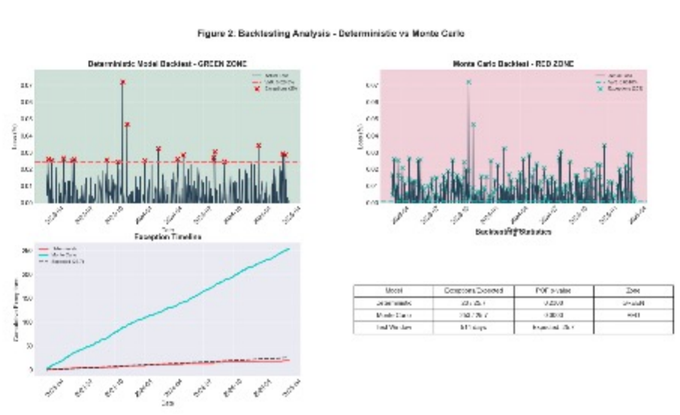
Evaluation Metrics (M4)
VaR Estimate 95% Exceptions count
Anticipated exceptions (5% test data)
POF Test p-value (statistical validity)
Zone: Green/Yellow/Red.
Runtime (computational cost O(N×sims)

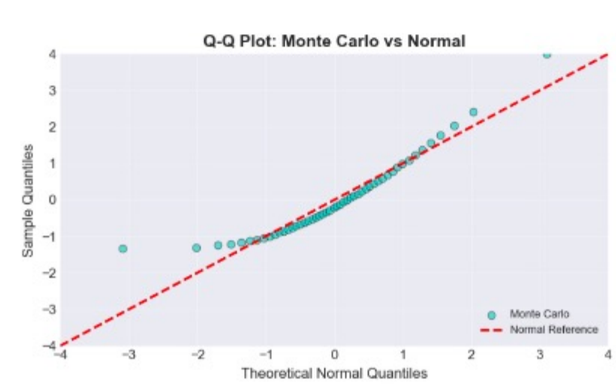
Comparative Results: Deterministic vs Monte Carlo

Metric	Deterministic	Monte Carlo
VaR Threshold	0.0243%	0.0010%
Exceptions / 514 days	20	253
POF p-value	0.2308	0.0000
Basel Zone	GREEN	RED (conservative)

Key Insights & Challenges
Q-Q plot: MC tails depart normality.
Histogram: wider spread than bell curve
MC too sensitive → lands in RED zone.
RED = overly conservative, not failure.
Day-independence simplification remains







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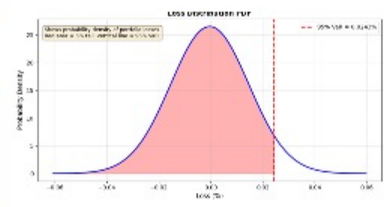

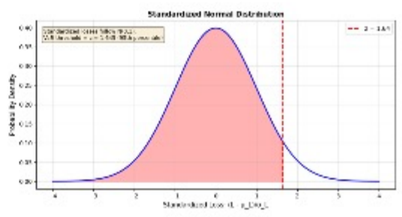

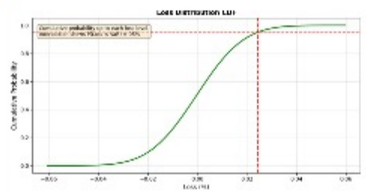

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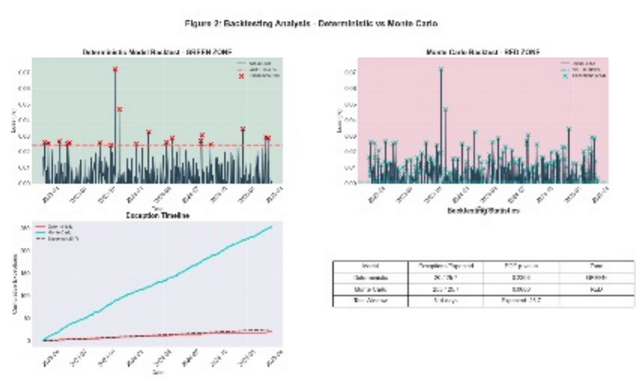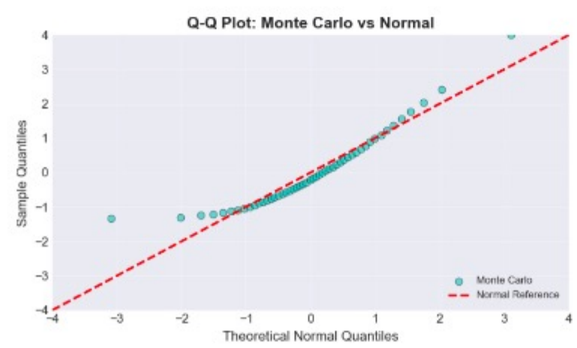
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